DAY 16 - INTERMEDIATE

Pivot Tables

Primary tool: Google Sheets

What you are learning

Fast grouped summaries and validation of grand totals.

Learning objectives

- Create a Pivot Table

- Choose rows/columns/values

- Validate grand totals

Worked practical

Summarize actuals by county and activity, then compare to source totals.

Source / Input	Method	Output / Check
Raw data	Apply the lesson skill	Result linked to source
Change one input	Recalculate / rerun	Confirm expected change
Control total / test	Validate	Explain the result

Guided practice

1. Recreate the lesson idea on a small example.
2. Change inputs and predict the result before recalculating.
3. Complete the supplied independent dataset in the required tool.
4. Create a control check or reconciliation.
5. Explain the result in ordinary language.

AI coaching

Ask AI to suggest a pivot layout for a stated question; build and verify it yourself.

AI discipline

Attempt first. Use AI for explanation, hints, debugging, critique, test cases and mock questioning. Never submit output you cannot explain and independently verify.

Independent exercise

Create operational Pivot Tables.

Common failure modes

- Wrong grain or key.
- Formula/query/code answers a different question.
- Missing or unmatched records are silently ignored.
- Findings are not reconciled to the numbers.
- AI-generated logic is accepted without testing.

Mastery gate

[ ] I can reproduce the method without copying.
[ ] I can trace the result to the source.
[ ] I have a QA/control check.
[ ] I can explain a limitation.
[ ] I can answer an assessor's follow-up question without asking AI.